

CN Lab 3 Tasks

1) Server

```
Work 1:CN_Lab_Tasks 2:FAST-Labs-KHI arch-host
1 import socket as s
2
3
4 def add(n1, n2):
5     return n1 + n2
6
7
8 def subtract(n1, n2):
9     return n1 - n2
10
11
12 def multiply(n1, n2):
13     return n1 * n2
14
15
16 def divide(n1, n2):
17     if n2 == 0:
18         return None
19     return n1 / n2
20
21
22 sock = s.socket()
23
24 PORT = 3001
25 sock.bind(("localhost", PORT))
26
27 sock.listen(1)
28 print(f"Server is listening on port {PORT}...")
29
30 conn, addr = sock.accept()
31
32 conn.send("Enter a number: ".encode())
33 num1 = int(conn.recv(512).decode())
34
35 conn.send("Enter another number: ".encode())
36 num2 = int(conn.recv(512).decode())
37
38 selection_prompt = """
39 Select an operator:
40 Type 1 for addition
41 Type 2 for subtraction
42 Type 3 for multiplication
43 Type 4 for division
44 Type 0 to exit
45 > """
46 conn.send(selection_prompt.encode())
47 selection = int(conn.recv(512).decode())
48
49 response = ""
50 match selection:
51     case 1:
52         res = add(num1, num2)
53         response = f"Result: {res}"
54     case 2:
55         res = subtract(num1, num2)
56         response = f"Result: {res}"
57     case 3:
58         res = multiply(num1, num2)
59         response = f"Result: {res}"
60     case 4:
61         res = divide(num1, num2)
62         if res is None:
63             response = "Error Division by zero is not allowed"
64         else:
65             response = f"Result: {res}"
66     case 0:
67         response = "Exiting..."
68     case _:
69         response = "Unknown Case Reached! Exiting..."
70
71 conn.send(response.encode())
72
73 print("Connection Closed.")
74 conn.close()
75 sock.close()
```

Client

```
Work 1:CN_Lab_Tasks 2:FAST-Labs-KHI arch-host
1 import socket as s
2
3 sock = s.socket()
4
5 PORT = 3001
6 sock.connect(("localhost", PORT))
7
8 prompt = sock.recv(512).decode()
9 while True:
10     try:
11         number = int(input(prompt))
12         break
13     except ValueError:
14         print("Value needed is any valid number. Try again.")
15
16 sock.send(str(number).encode())
17
18 prompt = sock.recv(512).decode()
19 while True:
20     try:
21         number = int(input(prompt))
22         break
23     except ValueError:
24         print("Value needed is any valid number. Try again.")
25
26 sock.send(str(number).encode())
27
28 prompt = sock.recv(512).decode()
29 while True:
30     try:
31         number = int(input(prompt))
32         if number in [1, 2, 3, 4, 0]:
33             break
34         else:
35             raise ValueError
36     except ValueError:
37         print("Enter a valid value. (0, 1, 2, 3, 4)")
38
39 sock.send(str(number).encode())
40
41 result = sock.recv(512).decode()
42 print(f"\n{result}")
43
44 sock.close()
```

```
NORMAL > main FAST-Labs-KHI > Lab3/.../task1/client.py ^[ < 6 Bot 44:12 17:19
```

Output

```
.../post_lab_tasks/task1 main ? } pytl  
Server is listening on port 3001...  
Connection Closed.
```

 Enable single-window sq

```
.../post_lab_tasks/task1 main ? }
```

```
.../post_lab_tasks/task1 main ? x python client.py
```

```
Enter a number: 15
```

```
Enter another number: 10
```

```
Select an operator:
```

```
Type 1 for addition
```

```
Type 2 for subtraction
```

```
Type 3 for multiplication
```

```
Type 4 for division
```

```
Type 0 to exit
```

```
> 3
```

```
Result: 150
```

```
.../post_lab_tasks/task1 main ? }
```

2) Server

```
Work 1:CN_Lab_Tasks 2:FAST-Labs-KHI arch-hos
1 import socket as s
2 import json
3
4 MAX_COURSES = 2
5
6 sock = s.socket()
7
8 PORT = 4321
9 sock.bind(("localhost", PORT))
10
11 sock.listen(1)
12 print(f"Server up and listening on {PORT}...")
13
14 conn, addr = sock.accept()
15 print(f"Client connected at {addr[0]}:{addr[1]}!")
16
17 try:
18     data = conn.recv(1024).decode()
19     if data:
20         new_data = json.loads(data)
21         try:
22             with open("db.json", "r") as file:
23                 db = json.load(file)
24         except (FileNotFoundError, json.JSONDecodeError):
25             db = {"students": []}
26         db["students"].extend(new_data)
27
28         with open("db.json", "w") as file:
29             json.dump(db, file, indent=2)
30
31         response = json.dumps(db).encode()
32         conn.send(response)
33 except Exception as e:
34     print(f"--Error--\n{e}")
35
36 conn.close()
37 sock.close()
```

Client

```

Work 1:CN_Lab_Tasks 2:FAST-Labs-KHI arch-host
1 import socket as s
2 import json
3
4 MAX_COURSES = 2
5
6
7 class Student:
8     def __init__(self, name):
9         self.name = name
10        self.marks = {}
11        self.total = 0
12        self.percentage = 0
13
14    def get_marks(self):
15        print("--Marks Registration--")
16        for i in range(MAX_COURSES):
17            sub = input("Subject Name: ")
18            mark = int(input("Subject Marks: "))
19            self.marks[sub] = mark
20        print(self.marks)
21
22    def calculate_total(self):
23        self.total = sum(self.marks.values())
24
25    def calculate_percentage(self):
26        self.percentage = self.total / (MAX_COURSES * 100)
27
28    def json_format(self):
29        return {
30            "name": self.name,
31            "marks": self.marks,
32            "total": self.total,
33            "percentage": self.percentage,
34        }
35
36 student_list = []
37 num_students = int(input("Enter number of students to log: "))
38 for i in range(num_students):
39     name = input("Enter name of student: ")
40     student = Student(name)
41     student.get_marks()
42     student.calculate_total()
43     student.calculate_percentage()
44     student_list.append(student)
45
46 sock = s.socket()
47
48 PORT = 4321
49 sock.connect(("localhost", PORT))
50
51 data = json.dumps([student.json_format() for student in student_list])
52 sock.send(data.encode())
53
54 res = json.loads(sock.recv(1024).decode())
55 print(res)
56
57 sock.close()

```

Output

```

~/post_lab_tasks/task2 main ? * python server.py
Server up and listening on 4321...
Client connected at 127.0.0.1:42632!

~/post_lab_tasks/task2 main ? )

~/post_lab_tasks/task2 main ? ) python client.py
Enter number of students to log: 1
Enter name of student: Aly2
--Marks Registration--
Subject Name: CN
Subject Marks: 100
Subject Name: Calc
Subject Marks: 82
{'CN': 100, 'Calc': 82}
{'students': [{'name': 'Aly', 'marks': {'DS': 89, 'DB': 98}, 'total': 187, 'percentage': 0.935}, {'name': 'Aly2', 'marks': {'CN': 100, 'Calc': 82}, 'total': 182, 'percentage': 0.911}]}

~/post_lab_tasks/task2 main ? )

```

3)

```
Work 1:task3 2:FAST-Labs-KHI arch-host
server.py x client.py x
1 import socket as s
2
3 sock = s.socket()
4
5 PORT = 3001
6 sock.bind(("localhost", PORT))
7
8 sock.listen(1)
9 print(f"Server is up and running at port {PORT}")
10
11 conn, addr = sock.accept()
12 print(f"Client connected at {addr[0]}:{addr[1]}!")
13
14 msg = conn.recv(512).decode()
15 print(f"Message from client: {msg}")
16
17 conn.close()
18 sock.close()

1 import socket as s
2
3 sock = s.socket()
4
5 PORT = 3001
6 sock.connect(("localhost", PORT))
7 print("Connected to Department Message Channel!")
8
9 msg = input("Enter Message for department: ")
10 sock.send(msg.encode())
11
12 sock.close()

NORHAL y main FAST-Labs-KHI Lab3/~/task3/client.py o 5 Bot 12:11 15:25

~/post_lab_tasks/task3 main ? ) python server.py
Server is up and running at port 3001
Client connected at 127.0.0.1:52726!
Message from client: Hello!

~/post_lab_tasks/task3 main ? )

~/post_lab_tasks/task3 main ? ) python client.py
Connected to Department Message Channel!
Enter Message for department: Hello!

~/post_lab_tasks/task3 main ? )
```

4)

Work

1:task4 2:FAST-Labs-KHI

arch-host

server.py x | client.py x

```

1 import json
2 import socket as s
3
4 sock = s.socket()
5
6 PORT = 3001
7 sock.bind(("localhost", PORT))
8
9 sock.listen(1)
10 print(f"Server is listening at port {PORT}...")
11
12 conn, addr = sock.accept()
13 print(f"Client connected at {addr[0]}:{addr[1]}!")
14
15 data = conn.recv(1024).decode()
16 student = json.loads(data)
17 print(f"Student ID: {student['id']}")
18 print(f"Student Name: {student['name']}")
19 print(f"Department: {student['department']}")
20
21 conn.send("Recieved!".encode())
22
23 conn.close()
24 sock.close()

```

```

1 import json
2 import socket as s
3
4 sock = s.socket()
5
6 PORT = 3001
7 sock.connect(("localhost", PORT))
8
9 print("--Student Information--")
10 id = input("Student ID: ")
11 name = input("Student Name: ")
12 dept = input("Department: ")
13
14 data = json.dumps({"id": id, "name": name, "department": dept})
15 sock.send(data.encode())
16
17 print(f"Server: {sock.recv(512).decode()}")
18
19 sock.close()

```

NORMAL

/ main

FAST-Labs-KHI

Lab3/./task4/server.py

ciw 6 75% 18:36 15:58

```

~/post_lab_tasks/task4 main ? ) python server.py
Server is listening at port 3001...
Client connected at 127.0.0.1:50936!
Student ID: 0512
Student Name: Aly
Department: CS
~/post_lab_tasks/task4 main ? )

```

```

~/post_lab_tasks/task4 main ? x python client.py
--Student Information--
Student ID: 0512
Student Name: Aly
Department: CS
Server: Recieved!
~/post_lab_tasks/task4 main ? )

```

5)

```
Work 1:task5 2:FAST-Labs-KHI arch-host
1 import socket as s
2
3 ports = [17, 21, 22, 25, 53, 69, 80, 107, 443, 6379]
4
5 print(f"PORT\tTCP Service\tUDP Service\n{'-' * 40}")
6
7 for port in ports:
8     try:
9         tcp_serv = s.getservbyport(port, "tcp")
10    except Exception:
11        tcp_serv = "Unknown"
12
13    try:
14        udp_serv = s.getservbyport(port, "udp")
15    except Exception:
16        udp_serv = "Unknown"
17
18    print(f"{port}\t{tcp_serv}\t{udp_serv}")
```

NORMAL ▶ main FAST-Labs-KHI > Lab3/~/task5/task.py ^Wl < 6 27% 5:43 16:07

~/post_lab_tasks/task5 main ?) python task.py

PORT	TCP Service	UDP Service
------	-------------	-------------

17	qotd	qotd
21	ftp	ftp
22	ssh	ssh
25	smtp	smtp
53	domain	domain
69	tftp	tftp
80	http	http
107	rftelnet	rftelnet
443	https	https
6379	redis	Unknown

~/post_lab_tasks/task5 main ?)

6) Server

```
Work 1:CN_Lab_Tasks 2:FAST-Labs-KHI arch-host
server.py x client.py x
2 import socket as s
3
4 letter_grades = {
5     "A+": 4.00,
6     "A-": 3.67,
7     "B+": 3.33,
8     "B-": 3.00,
9     "B-": 2.67,
10    "C+": 2.33,
11    "C-": 2.00,
12    "C-": 1.67,
13    "D+": 1.33,
14    "D-": 1.00,
15    "F-": 0.00,
16 }
17
18
19 class GPADData:
20     def __init__(self):
21         self.sgpa = 0
22         self.total_credithr = 0
23         self.total_quality_pts = 0
24
25     def calculate(self, data):
26         total_g = 0
27         for course in data:
28             total_g += letter_grades[course["grade"]]
29             self.total_credithr += course["hours"]
30             course["quality_points"] = course["hours"] * letter_grades[course["grade"]]
31             self.total_quality_pts += course["quality_points"]
32         self.sgpa = float(self.total_quality_pts / self.total_credithr)
33
34     def generate_report(self, data):
35         report = f"Course\tCH\tGrade\tGPA\tQuality Points\n('-' * 50)\n"
36
37         for course in data:
38             report += f"{course['name']}\t\t{course['hours']}\t\t{course['grade']}\t\t{letter_grades[course['grade']]}\n"
39             report += f"{'-' * 50}\n"
40             report += f"Total\t\t{self.total_credithr}\t\t\t\t{self.total_quality_pts:.2f}\n"
41             report += f"Semester GPA: {self.sgpa:.2f}"
42
43         return report
44
45
46 sock = s.socket()
47
48 PORT = 3001
49 sock.bind(("localhost", PORT))
50
51 sock.listen(1)
52 print(f"Server is up and running at port {PORT}")
53
54 conn, addr = sock.accept()
55 print(f"Client connected at {addr[0]}:{addr[1]}")
56
57 msg = conn.recv(1024).decode()
58 semester = json.loads(msg)
59
60 gpa = GPADData()
61
62 gpa.calculate(semester)
63 report = gpa.generate_report(semester)
64
65 conn.send(report.encode())
66
67 conn.close()
68 sock.close()
```

Client

```
Work 1:CN_Lab_Tasks 2:FAST-Labs-KHI arch-host
server.py x client.py x
1 import json
2 import socket as s
3
4 MAX_COURSES = 2
5
6 sock = s.socket()
7
8 PORT = 3001
9 sock.connect(("localhost", PORT))
10
11 courses = []
12 for i in range(MAX_COURSES):
13     course = {}
14     print(f"Course {i + 1}")
15     course["name"] = input("Enter Course Name: ")
16     course["hours"] = int(input("Enter Course Hours: "))
17     course["grade"] = input("Enter Letter Grade: ")
18     courses.append(course)
19
20 msg = json.dumps(courses)
21
22 sock.send(msg.encode())
23
24 report = sock.recv(1024).decode()
25 print(f"--Report--\n{report}")
26
27 sock.close()
```

NORMAL main FAST-Labs-KHI Lab3/.../task6/client.py gj 6 7% 2:18 17:26

Output

```
.../post_lab_tasks/task6 main ? ) python server.py
Server is up and running at port 3001
Client connected at 127.0.0.1:41132!
```

```
.../post_lab_tasks/task6 main ? )
```

```
.../post_lab_tasks/task6 main ? ) python client.py
```

Course 1

Enter Course Name: OS

Enter Course Hours: 3

Enter Letter Grade: A

Course 2

Enter Course Name: DB

Enter Course Hours: 3

Enter Letter Grade: B+

--Report--

Course	CH	Grade	GP	Quality Points
OS	3	A	4.0	12.0
DB	3	B+	3.33	9.99
Total				21.99
Semester GPA:				3.67

```
.../post_lab_tasks/task6 main ? )
```

7)

```
Work 1:task7 2:FAST-Labs-KHI arch-host
server.py client.py x
1 import json
2 import socket as s
3
4
5 class StudentData:
6     def __init__(self):
7         self.filename = "student.json"
8
9     def process_data(self, data):
10         with open(self.filename, "w") as f:
11             json.dump(data, f, indent=4)
12
13         with open(self.filename, "r") as f:
14             saved_data = json.load(f)
15
16         return saved_data
17
18     def generate_display(self, data):
19         display = f"Student ID:\t{data['student_id']}\n"
20         display += f"Name:\t{data['name']}\n"
21         display += f"Department:\t{data['department']}\n"
22         display += f"Semester:\t{data['semester']}\n"
23         display += f"GPA:\t{data['gpa']}\n"
24         return display
25
26
27 sock = s.socket()
28
29 PORT = 3001
30 sock.bind(("localhost", PORT))
31
32 sock.listen(1)
33 print(f"Server is up and running at port {PORT}")
34
35 conn, addr = sock.accept()
36 print(f"Client connected at {addr[0]}:{addr[1]}")
37
38 msg = conn.recv(1024).decode()
39 student = json.loads(msg)
40
41 st_data = StudentData()
42
43 saved_student = st_data.process_data(student)
44 display_msg = st_data.generate_display(saved_student)
45
46 conn.send(display_msg.encode())
47
48 conn.close()
49 sock.close()
50
51
52 import json
53 import socket as s
54
55 sock = s.socket()
56
57 PORT = 3001
58 sock.connect(("localhost", PORT))
59
60 student = {}
61 print("Enter Student Information")
62 student["student_id"] = input("Enter Student ID: ")
63 student["name"] = input("Enter Name: ")
64 student["department"] = input("Enter Department: ")
65 student["semester"] = int(input("Enter Semester: "))
66 student["gpa"] = float(input("Enter GPA: "))
67
68 msg = json.dumps(student)
69
70 sock.send(msg.encode())
71
72 display = sock.recv(1024).decode()
73 print(f"\n--Student Information--\n{display}")
74
75 sock.close()
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